Lecture 20: Flux Shaping and Power Peaking (The "Hot Spot" Problem)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: The "Average" Trap

In previous lectures, we treated the reactor as a "Point" (k_{eff}) or a perfect "Sine Wave." In reality, a reactor core is a heterogeneous mix of fuel, poison, water, and steel.

The Problem: The safety of a reactor is limited by its **hottest** fuel pin, not its average pin.

- If the reactor is producing 3000 MW total, but one region is producing 5x the average power density, that region will melt.
- We define the **Peaking Factor** (F_q):

$$F_q = \frac{\text{Maximum Power Density}}{\text{Average Power Density}}$$

Goal: The goal of reactor design is to make F_q as close to 1.0 as possible (i.e., "Flatten the Flux").

1 The Natural Shape: The Cosine

For a uniform, bare cylindrical reactor, diffusion theory dictates the flux shape is a Bessel function (radial) and Cosine (axial):

$$\phi(r, z) = \phi_{max} J_0\left(\frac{2.405r}{R}\right) \cos\left(\frac{\pi z}{H}\right)$$

To see just how bad this "natural" shape is for engineering purposes, we can break the total peaking factor (F_q) into its radial (F_r) and axial (F_z) components.

Integrating the cosine wave over the height of the core gives an axial peaking factor of:

$$F_z = \frac{\pi}{2} \approx 1.57$$

Integrating the Bessel function over the radius of the core gives a radial peaking factor of:

$$F_r \approx 2.32$$

Therefore, the total core peaking factor for a bare cylinder is:

$$F_q = F_r \times F_z = 1.57 \times 2.32 \approx 3.64$$

This is terrible! It means the center of the core is working nearly 4 times harder than the edges. We are wasting the capacity of the fuel at the periphery.

2 Reflector Savings

Surrounding the core with a reflector (water, graphite, or beryllium) is the most fundamental way to improve neutron economy and instantly flatten the power distribution.

2.1 How a Reflector Works

A reflector is not a "mirror" in the optical sense. It works via **diffusive scattering**.

- Neutrons leak out of the fuel region into the reflector material.
- Instead of escaping to the environment, they collide with the reflector nuclei.
- The reflector acts as a "random walk" generator. A significant fraction of these neutrons (measured by the **Albedo** or reflection coefficient, β) scatter back into the core.
- **The Result:** The neutron flux at the core boundary (R) is no longer zero. It is raised to some finite value.
- Mathematically, this makes the core "feel" like it is larger than it physically is. We call this effective increase the **Reflector Savings** (δ).

  $$R_{bare} = R_{physical} + \delta$$

  Because the flux does not have to drop to zero at the physical edge, the cosine shape in the center is flattened, and F_q drops from 3.64 to ≈ 2.5.

2.2 Material Selection: Who Uses What and Why?

A good reflector must have a high scattering cross-section (Σ_s) and a low absorption cross-section (Σ_a). However, real-world material selection is dictated by cost, physical size limits, and the specific goals of the reactor.

- **Light Water (H_2O) — The Commercial Standard**
  - **Who uses it:** Every commercial PWR and BWR.
  - **Why:** It is incredibly cheap, fluid, and already surrounds the core as the primary coolant and moderator. Hydrogen is exceptionally light, meaning neutrons lose massive amounts of energy per collision (excellent thermalization).
  - **The Catch:** Hydrogen has a measurable absorption cross-section (forming Deuterium). It is a "good enough" reflector for commercial use, but not the most neutron-efficient.
- **Graphite (C) — The High-Temperature & Legacy Reflector**
  - **Who uses it:** Gas-cooled reactors (HTGRs, AGRs) and legacy Soviet designs (RBMK).
  - **Why:** Carbon has an extremely low absorption cross-section compared to water, making for excellent neutron economy. It also maintains its structural integrity at staggeringly high temperatures, which is required for gas-cooled reactors.
  - **The Catch:** Because Carbon is heavier than Hydrogen, it takes more collisions to thermalize a neutron. To be an effective reflector, a graphite blanket must be physically massive (often meters thick), which is impractical for most modern pressure vessel designs.

- **Beryllium (Be) — The "Gold Standard" for Research & Space**
  - **Who uses it:** High-flux research reactors (e.g., HFIR at Oak Ridge), space microreactors (NASA's Kilopower), and nuclear weapons.
  - **Why:** It is lightweight and has very low absorption. More importantly, it actually *multiplies* neutrons via the $(n, 2n)$ reaction:

$$^{9}\mathrm{Be} + \mathrm{n} \rightarrow 2\alpha + 2\mathrm{n}$$

  - **The Catch:** Beryllium is staggeringly expensive, difficult to machine, and highly toxic (causing berylliosis). Furthermore, the (n, α) reactions build up Helium and Tritium gas inside the metal lattice over time, causing the reflector blocks to physically swell and crack, requiring complete replacement every decade.
- **Stainless Steel — The Modern "Heavy Reflector"**
  - **Who uses it:** Generation III+ commercial reactors (like the Westinghouse AP1000 and the French EPR).
  - **Why:** While steel has a higher thermal neutron absorption cross-section than water, it is an excellent *fast* neutron reflector. Designers replaced the water gaps at the edge of the core with a massive, solid steel cylinder. This heavy reflector bounces fast neutrons back into the outer peripheral fuel assemblies (boosting their power output) while acting as a massive shield to protect the reactor pressure vessel from fast neutron embrittlement.

3 Sources of Flux Tilting and Spatial Instability

Even with reflectors, operational changes can severely distort the flux profile.

3.1 Distortion by Control Rods (Flux Tilting)

Inserting control rods distorts the natural shape, poking "holes" in the neutron flux.

- **Partial Insertion:** If we insert a rod bank halfway ($z = H/2$), the flux is pushed to the bottom of the core. The peak shifts and intensifies.
- **Asymmetry:** If we insert rods on the left but not the right, the flux "tilts" to the right.

This creates local zones where $k_\infty > 1$ feeding zones where $k_\infty < 1$.

3.2 The Consequence of Size: Spatial Instability (Xenon)

If a reactor is physically very large (specifically, large compared to the neutron **Migration Length**, M), the regions become "decoupled." A perturbation in the top half may not be instantly "felt" by the bottom half.

This decoupling allows for massive, slow-motion power swings driven by fission product poisons, primarily **Xenon-135**.

- **The Delay Mechanism:** Xenon-135 is a massive neutron poison, but it is not produced instantly. It is the decay product of Iodine-135 (which has a half-life of 6.6 hours). Xenon-135 then decays away with a half-life of 9.2 hours.

- **The Pendulum Effect (Xenon Oscillations):**
  1. Power inadvertently shifts to the top of the core.
  2. Fission increases there, producing lots of Iodine-135.
  3. Hours later, that Iodine decays into a massive cloud of Xenon poison at the top of the core.
  4. The Xenon chokes out the chain reaction at the top, pushing the power aggressively to the bottom of the core.
  5. Hours later, the Xenon burns out at the top, while a new Xenon cloud forms at the bottom. Power swings back to the top.
- Because of the half-lives involved, this creates a 24-to-30 hour power oscillation. Operators effectively end up "chasing" the power bubble around the core with partial control rod insertions.

The Chernobyl Connection (RBMK Design)
The RBMK-1000 was a massive reactor (7m high, 12m diameter). Its large graphite moderator made the Migration Length short relative to the core size. It was highly susceptible to "decoupled" spatial power instabilities and severe Xenon oscillations, requiring complex automatic control systems that were disastrously disabled on the night of the accident.

4 Chemical Shim: Eliminating Rod Distortion

To avoid the flux tilting issues caused by control rods, modern Pressurized Water Reactors (PWRs) use **Chemical Shim**. We dissolve Boric Acid directly into the primary coolant.

Benefit: The poison is distributed uniformly in the liquid, depressing the flux everywhere equally. This allows the reactor to run with "All Rods Out" (ARO), preserving the natural, smoother flux shape. However, this introduces complex chemical challenges.

4.1 Acid Neutralization and Lithium-7

Boric acid drops the pH of the primary coolant, which would cause severe corrosion of the steel vessel and Zircaloy cladding at 300°C. To buffer this, operators continuously inject Lithium Hydroxide (LiOH).

The Tritium Problem: Natural lithium contains $\approx 7.5\%$ Lithium-6, which has a massive absorption cross-section (≈ 940 barns) and reacts to form Tritium:

$$^{6}\text{Li} + \text{n} \rightarrow ^{3}\text{H} + \alpha$$

Tritium is a highly mobile radioactive gas that easily permeates pipes. Therefore, to neutralize the boric acid, PWRs must use highly enriched **Lithium-7** ($> 99.9\%$).

4.2 CRUD and CIPS

In the hottest regions of the core, localized subcooled boiling can occur. When the water turns to steam on the fuel rod surface, it leaves dissolved Boron and Lithium behind, baked onto the cladding as a crystalline deposit.

- This is known in the industry as **CRUD** (Chalk River Unidentified Deposit).
- Because this crud contains Boron, it creates an artificial, localized neutron poison. This suppresses power in the top of the core and pushes it to the bottom, a phenomenon known as **Crud Induced Power Shift (CIPS)**.

Historical Note: The Origin of "CRUD"
In the early 1950s, researchers operating the National Research Experimental (NRX) reactor at the Chalk River Laboratories in Canada pulled water-cooled fuel test assemblies out of the core and found them coated in a dark, highly radioactive crystalline scale. Because primary coolant analytical chemistry was still in its infancy, they didn't know the exact composition. Engineers simply logged it as a **"Chalk River Unidentified Deposit"** (CRUD).

The name stuck permanently. Today, even though we know exactly what the deposit is (primarily nickel ferrites and iron oxides baked onto the Zircaloy cladding), the entire nuclear navy and civilian power industry still calls it CRUD. In modern regulatory documents, you will occasionally see it backronymed to "Corrosion Related Unidentified Deposit" to sound more generic.

4.3 The Economics of Isotopic Enrichment: Boron-10 and Lithium-7

While Chemical Shim successfully flattens the flux without rod distortion, it introduces a complex economic and chemical balancing act regarding isotopic enrichment.

- **The Cost of Lithium-7:** Separating ^{7}Li from natural lithium is inherently difficult, historically requiring highly toxic mercury amalgam processes. Consequently, the high-purity ^{7}Li required by PWRs to neutralize boric acid without producing Tritium is staggeringly expensive and highly supply-chain constrained.
- **Enriched Boric Acid (EBA):** Natural boron is cheap, but it is only 20% Boron-10 (the active neutron absorber) and 80% Boron-11 (dead weight that only adds acidity to the water). To reduce the need for expensive ^{7}Li neutralization, modern plants increasingly pay a premium for Enriched Boric Acid (often enriched to $>$ 40% B-10).
- **The Economic Trade-Off:** While EBA is far more expensive upfront than natural boric acid, the cascading operational savings are massive:
  1. **Direct Chemical Savings:** Less total acid in the primary loop means significantly less of the hyper-expensive ^{7}Li is required to maintain a pH of 7.2.
  2. **Avoiding Derates (CIPS):** Lowering the total concentration of dissolved chemicals (both Boron and Lithium) starves the CRUD formation process. Avoiding Crud Induced Power Shift (CIPS) means the reactor does not have to be forced into a "derate" (running at lower power to stay within thermal limits), saving millions of dollars in lost electricity revenue.
  3. **Enabling 24-Month Cycles:** EBA allows plants to hold down the massive initial reactivity needed for a 24-month fuel cycle without violating maximum pH coefficient limits. Fewer refueling outages drastically increase the plant's lifetime capacity factor and profitability.

4.4 The Safety Limit: The Moderator Temperature Coefficient (MTC)

While chemical shim is incredibly useful for flux flattening, its use is strictly limited by reactor safety physics, specifically the Moderator Temperature Coefficient (MTC).

- **The Natural Safety Loop (Negative MTC):** Normally, as a reactor heats up, the water coolant expands (density decreases). Less dense water means less moderation, which naturally slows the chain reaction.
- **The Boron Reversal:** When the water is laced with soluble boron, expanding water pushes both moderator *and* poison out of the core. Removing poison **adds** reactivity.
- **The BOL Danger:** At the Beginning of Life (BOL), if the required boron concentration is too high, the positive reactivity added by the exiting boron outweighs the negative reactivity of the exiting moderator. This creates a **Positive MTC**—meaning if the reactor gets hotter, power goes *up* in a runaway feedback loop.
- **Why Enriched Boric Acid (EBA) Doesn't Help:** EBA solves the chemical pH and CRUD problems, but it provides zero benefit to the MTC problem. Reactivity hold-down requires a fixed total number of Boron-10 atoms in the core volume. When water expands, it pushes the exact same number of Boron-10 atoms out of the core regardless of whether natural or enriched acid was used.
- **The Solution:** NRC regulations strictly cap the maximum worth of soluble boron to ensure the MTC remains negative. To hold down the remaining excess reactivity at BOL without violating this limit, reactors are forced to rely on solid Burnable Poisons (IFBA/WABA) integrated directly into the fuel matrix.

5 Core Loading Patterns: The Ultimate Compromise

Even with chemical shim, we still need to manage radial flux peaking. We do this by adjusting the enrichment loading pattern of the fuel assemblies.

5.1 The Textbook Approach (Out-In Loading)

To get a perfectly flat flux ($F_q = 1.0$), you would logically load **lower** enrichment in the center and **higher** enrichment on the periphery. This boosts the "lazy" edges and suppresses the center peak.

- **The Fatal Flaw:** Putting the hottest, most highly enriched fuel on the outside of the core means a massive flux of fast neutrons hits the steel Reactor Pressure Vessel (RPV) directly. This causes rapid **neutron embrittlement** and drastically decreases the lifespan of the multibillion-dollar vessel.

5.2 The Real Solution: Low-Leakage Cores

To save the steel vessel, modern reactors do the exact opposite of the textbook ideal:

- **The Sacrificial Shield:** The oldest, most depleted "dead" fuel (often on its third fuel cycle) is placed on the outer periphery. It generates very little power, but acts as a highly effective buffer to absorb neutrons and shield the RPV wall.
- **The Hot Center:** The fresh, highly enriched fuel is pushed to the interior of the core.

5.3 Burnable Poisons (Fixing the Center Peak)

Pushing all the fresh fuel into a smaller central volume creates a massive, unsafe power peak. To prevent the center from melting, we must heavily poison the fresh fuel assemblies using solid **Burnable Poisons**.

- **IFBA (Integral Fuel Burnable Absorber):** A microscopic coating of Zirconium Diboride (ZrB_2) painted onto the UO_2 fuel pellets.
- **WABA (Wet Annular Burnable Absorber):** Hollow tubes of Boron Carbide inserted directly into the assembly guide thimbles.
- These poisons hold down the center peak early in life and burn away at the same rate the Uranium depletes, keeping the power profile flat over an 18-month cycle.

6 Summary

- F_q **(Peaking Factor):** The ratio of Max to Average power. Left alone, the bare core F_q is unacceptably high (≈ 3.64).
- **Reflectors:** Scatter neutrons back to the core, raising edge flux and flattening the center.
- **Xenon Oscillations & Rod Tilting:** Using rods pokes holes in the flux, exacerbating massive spatial power swings in large cores.
- **Chemical Shim:** Uses uniform liquid poison (Boron) neutralized by ^{7}Li to maintain ARO (All Rods Out) operation, though it can cause CIPS (CRUD).
- **Low-Leakage Loading:** We use depleted fuel on the outside to shield the vessel, and use solid burnable poisons (IFBA/WABA) to artificially suppress the hot fresh fuel in the center.

7 Case Study: The Naval Reactor (The "Life-of-Ship" Core)

Naval reactors operate under a completely different set of engineering constraints. They must fit in a confined space, survive combat shock, rapidly change power levels for tactical maneuvering, and operate for 25 to 33 years without ever being refueled.

To achieve a "life-of-ship" core, the reactor is loaded with Highly Enriched Uranium (HEU). This means the core has a staggering amount of excess reactivity at Beginning of Life (BOL) that must be safely held down and shaped.

- **No Chemical Shim:** Naval reactors generally cannot use soluble boron for daily reactivity control. In a combat scenario, a ship might need to go from 10% power to 100% power in seconds or minutes. You cannot pump fluid into and out of the primary loop fast enough to support that kind of rapid, extreme transient.
- **Heavy Reliance on Solid Poisons:** Because they cannot use liquid boron, naval cores rely intensely on solid burnable poisons integrated directly into the core matrix. These poisons are meticulously zoned to hold down the massive BOL excess reactivity of the HEU and to flatten the flux over a multi-decade lifespan.

- **Hafnium Control Rods:** Instead of Boron Carbide or Silver-Indium-Cadmium used in civilian plants, naval reactors utilize **Hafnium** for their control rods. Hafnium is a remarkable neutron absorber because its isotopes absorb neutrons and transmute into *other* Hafnium isotopes that are also excellent neutron absorbers ($^{177}\text{Hf} \rightarrow^{178} \text{Hf} \rightarrow^{179} \text{Hf}$). This allows the control rods to endure decades of neutron bombardment without burning out or losing their worth.

- **The Water Reflector vs. The Shield Tank:** The primary reflector in a naval reactor is the pressurized light water coolant circulating *inside* the thick steel RPV. While the reactor vessel itself is surrounded by a large tank of water (the primary shield tank), this outer pool acts almost exclusively as a biological shield to protect the crew from neutron and gamma radiation. Neutrons that leak through the thick steel RPV into the shield tank do not scatter back into the core.

References

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